

Unit 0 Project Brief

Aerospace Readiness Mini-Challenge

Mission Readiness: Safety, Tools & Engineering Systems

Challenge Statement: Design, build, test, and document a small aerospace support system that solves a simple classroom mission need while following all safety, certification, documentation, and teamwork expectations.

Project Overview

In this launch project, your team demonstrates that you are ready to work safely, document engineering decisions, use shared FabLab tools responsibly, and think about aerospace systems as connected subsystems. The build is intentionally small; the focus is on readiness, process, and evidence.

Mission Scenario

Aerospace teams do not begin major missions by building the largest system first. They verify procedures, roles, tools, communication, and readiness through smaller tasks before moving into more complex work. Your team will complete a mini-challenge that proves you are prepared for future POE projects.

Design Requirements

- Use an approved aerospace support problem such as a drone landing marker, payload fixture, rocket component stand, VEX test base, or motion-test fixture.
- Complete or confirm required shared FabLab certifications before using controlled tools independently.
- Create a clear problem statement, criteria, constraints, and at least one labeled concept sketch.
- Use safe tool habits, correct PPE, and clean workspace procedures throughout the activity.
- Build a small physical prototype or fixture using approved classroom materials.
- Test whether the system meets its intended purpose and document at least one improvement recommendation.

Constraints

- Project must stay small enough to complete during the Unit 0 readiness window.
- Only approved materials and tools may be used.
- No independent laser, 3D printer, or hand tool use until the correct certification is complete.
- Prototype must not create sharp edges, unstable supports, unsafe moving parts, or unsafe drone/rocket conditions.
- Final evidence must be documented in the engineering notebook or assigned digital format.

Required Engineering Evidence

Problem Definition	Problem statement, criteria, constraints, and selected mini-challenge.
Design Evidence	Labeled sketch, planned materials, and explanation of how the system is supposed to work.
Safety Evidence	Certification status, PPE/tool notes, and safe work habits.
Testing Evidence	Photo, observation notes, simple measurement, pass/fail check, or user feedback.

Reflection	What worked, what failed, and what you would improve before a larger aerospace project.
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Project Checkpoints

- 1 Certifications confirmed
- 2 Problem and sketch approved
- 3 Prototype built safely
- 4 Test completed
- 5 Final readiness reflection submitted

Success Criteria

Readiness & Safety	Certifications, PPE, tool habits, and workspace expectations are followed consistently.
Design Thinking	Problem, criteria, constraints, and concept are clear and connected to an aerospace need.
Prototype & Test	Prototype is built safely and tested against its intended purpose.
Documentation	Notebook evidence is complete, organized, and useful for future engineering work.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.

Unit 1 Project Brief

VEX Aerospace Mechanisms Challenge

Mechanisms of Flight: Energy, Power & Aerospace Machines

Challenge Statement: Design and test a VEX mechanism that performs an aerospace support task by redirecting energy and motion in a measurable way.

Project Overview

In this project, your team uses VEX parts to design a mechanism that changes force, speed, direction, distance, or motion for an aerospace support task. Your final design must be supported by calculations, prototype evidence, and repeatable test data.

Mission Scenario

Aerospace missions depend on mechanisms: launch support equipment, hatch doors, payload lifts, landing gear, antenna positioners, and deployment systems all use moving parts to transfer energy. Your team will create a VEX-based mechanism that performs one of these support functions and prove its performance with data.

Design Requirements

- Select an approved mission task such as payload lift, launch angle adjuster, deployable support, hatch/bay door, antenna or sensor positioner, or another approved aerospace mechanism.
- Use at least one mechanism type such as gears, pulleys, sprockets, levers, linkages, wheels, axles, or compound machines.
- Identify input motion, output motion, and how the mechanism changes force, speed, distance, direction, or torque.
- Calculate relevant values such as mechanical advantage, drive ratio, work, power, or efficiency.
- Build a stable VEX prototype with appropriate spacing, shaft support, alignment, and fastening.
- Run multiple trials and use data to judge performance and reliability.

Constraints

- Prototype must be primarily built from VEX parts and approved classroom materials.
- Mechanism must complete a measurable motion or load task.
- Design must be safe to operate and must not launch objects toward people or unsafe areas.
- Teams must document at least two design concepts before selecting one.
- Final presentation must include calculations and test data, not just a demonstration.

Required Engineering Evidence

Design Brief	Problem statement, mission task, criteria, constraints, and final performance goal.
Concept Development	Individual/team sketches, decision matrix, and selected mechanism rationale.
Calculations	Mechanical advantage, gear/drive ratio, work, power, efficiency, or other relevant values.
Testing	Repeated trials measuring load, distance, angle, time, success rate, reliability, or efficiency.

Design Review	Prototype photos, test data, final claim, evidence, and improvement recommendation.
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Project Checkpoints

- 1 Design brief interpreted
- 2 Concepts and decision matrix complete
- 3 Prototype build approved
- 4 Testing data collected
- 5 Final design review delivered

Success Criteria

Mechanism Function	Design performs the selected aerospace task in a measurable and repeatable way.
Engineering Analysis	Calculations correctly connect mechanism design to performance.
Build Quality	Prototype is stable, aligned, safe, and intentionally constructed.
Evidence-Based Defense	Final recommendation is supported by sketches, calculations, test data, and reflection.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.

Unit 2 Project Brief

Aerospace Payload Support Structure Challenge

Built to Survive: Aerospace Structures, Materials & Testing

Challenge Statement: Design, build, and test a lightweight payload support or aerospace structure that safely supports a required load while minimizing unnecessary mass.

Project Overview

In this project, your team designs and tests a lightweight aerospace structure that supports a required payload while managing mass, stiffness, stability, and material choice. Your design decisions must be backed by force diagrams, calculations, material reasoning, and physical test data.

Mission Scenario

Every aerospace structure must balance strength, weight, reliability, and material limits. Launch towers, payload frames, drone supports, rover chassis, aircraft structures, and satellite brackets must carry loads without wasting mass. Your team will design and test a structure that proves it can survive its mission.

Design Requirements

- Select an approved structure type such as payload support tower, launch support frame, drone landing support, truss, test stand, or aerospace bracket/frame model.
- Create free body diagrams or force diagrams showing the major external forces on the structure.
- Use structural ideas such as tension, compression, bending, truss geometry, centroid, moment, stiffness, or load path.
- Choose approved materials and explain why they are appropriate based on properties such as strength, mass, stiffness, density, or manufacturability.
- Build a prototype and test it using a controlled load procedure.
- Evaluate performance using load held, mass, deflection, structural efficiency, failure mode, or safety factor evidence.

Constraints

- Structure must fit within the approved size limits set for the classroom test area.
- Structure must use only approved materials, adhesives, fasteners, and fabrication methods.
- Load testing must follow the assigned safety procedure and may only happen in the approved test zone.
- Design must include a clear load path from payload/load point to support/base.
- Final design must be defended using both analysis and physical test evidence.

Required Engineering Evidence

Analysis	Force diagram, load path notes, moment/equilibrium/truss reasoning, or other assigned calculations.
Material Selection	Material properties, density/mass notes, and explanation of material choice.
Design Development	Concept sketches, decision matrix, and selected structure plan.

Testing	Load, mass, deflection, failure mode, and structural efficiency data.
Final Recommendation	Evidence-based conclusion about whether the structure met the mission requirements.

Project Checkpoints

- 1 Mission requirements identified
- 2 Force diagram and concept sketches complete
- 3 Material plan approved
- 4 Prototype built
- 5 Load test and analysis complete

Success Criteria

Structural Performance	Structure supports the required load safely and with clear stability.
Analysis & Calculations	Force, load path, and material reasoning support the design choices.
Testing Quality	Testing is repeatable, documented, and connected to the criteria.
Design Communication	Final review explains tradeoffs among strength, mass, stiffness, and failure risk.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.

Unit 3 Project Brief

Drone Mission + VEX Ground Support System

Autonomous Aerospace Systems: Controls, Drones & Fluid Power

Challenge Statement: Program and test an autonomous drone mission, then integrate a ground support system that helps the mission operate safely, accurately, or reliably.

Project Overview

In this project, your team programs a Robolink drone mission and designs a VEX, sensor-based, or fluid-power ground support system that connects to the mission environment. The focus is control logic, system behavior, feedback, reliability, and safe operation.

Mission Scenario

Modern aerospace systems combine air vehicles, ground support, sensors, automation, and control logic. A drone may inspect, deliver, survey, or land while a ground system guides, signals, positions, protects, or triggers part of the mission. Your team will create a small-scale version of that interaction.

Design Requirements

- Create a mission objective such as inspection, survey, target landing, obstacle navigation, payload simulation, or checkpoint route.
- Develop a flowchart before programming the mission.
- Program at least one open-loop mission behavior and identify its limitations.
- Add at least one control improvement using sensor feedback, conditional logic, thresholds, repeated testing, or revised timing.
- Design a ground support element such as landing pad, gate, beacon, inspection zone, sensor target, payload platform, VEX actuator, or fluid-power model.
- Test mission performance across repeated trials and document reliability, accuracy, time, or success rate.

Constraints

- All drone operations must follow classroom flight safety rules and approved flight zones.
- Drone code must be tested progressively; unsafe or unpredictable code may not be flown.
- Ground support system must not contact propellers or create unsafe flight obstacles.
- VEX or fluid-power components must be stable and safe before they interact with a drone mission.
- Final mission must be documented with flowchart, code evidence, test data, and system diagram.

Required Engineering Evidence

Control Logic	Flowchart showing sequence, decisions, inputs, outputs, and feedback points.
Programming Evidence	Code screenshots or file evidence with comments explaining key mission logic.
System Integration	Diagram showing the drone, ground support system, inputs, outputs, sensors, and interactions.
Testing	Repeated trials measuring mission success, accuracy, time, reliability, or error.

Improvement	Explanation of how feedback, code revision, or ground support improved performance.
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Project Checkpoints

- 1 Mission objective approved
- 2 Flowchart complete
- 3 Open-loop test completed
- 4 Ground system integrated
- 5 Final mission test and review delivered

Success Criteria

Mission Function	Drone and ground support system complete the intended mission safely and repeatably.
Control System Thinking	Flowchart, code, sensors, feedback, and logic are clearly connected.
Integration	Ground support system meaningfully supports or interacts with the mission.
Testing & Debugging	Performance data and revisions show thoughtful debugging and improvement.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.

Unit 4 Project Brief

Flight Performance Data Investigation

Flight Performance: Statistics, Kinematics & Test Data

Challenge Statement: Design and conduct a controlled flight or motion investigation, collect repeated data, analyze variation, and use kinematic reasoning to explain performance.

Project Overview

In this investigation, your team uses repeated trials and motion data to evaluate how an aerospace object moves or performs. The focus is not just making something fly or move - it is using statistics, graphs, and kinematics to make a defensible performance claim.

Mission Scenario

Aerospace engineers rely on test data before trusting a system. Gliders, projectiles, drones, rovers, launch devices, and dropped objects all produce motion data that can be measured, graphed, compared, and used to improve design decisions. Your team will run a controlled investigation and defend a conclusion using evidence.

Design Requirements

- Select an approved investigation such as glider flight, projectile launch, drop test, drone route timing, rover motion, or launch angle/range study.
- Identify independent, dependent, and controlled variables.
- Collect repeated trial data under consistent testing conditions.
- Create at least one graph or visual representation of the data.
- Calculate appropriate statistics such as mean, median, range, standard deviation, percent error, success rate, or variation between trials.
- Use kinematic ideas such as distance, displacement, velocity, acceleration, free fall, projectile motion, range, or launch angle to explain results.

Constraints

- Test setup must be safe, repeatable, and approved before trials begin.
- Objects must not be launched, dropped, or flown toward people, fragile objects, or unsafe areas.
- Data must come from multiple trials, not a single best run.
- Only one primary variable should be changed at a time unless approved.
- Final claim must be based on data and calculations, not opinion or appearance.

Required Engineering Evidence

Investigation Plan	Research question, variables, procedure, equipment list, and safety notes.
Raw Data	Repeated trial data with units, notes, and any rejected-trial explanation.
Statistics	Mean, variation, range, standard deviation, success rate, or other assigned data analysis.
Kinematics	Motion calculations or graphs connecting performance to physics concepts.

Conclusion	Evidence-based claim about performance and a recommended improvement or next test.
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Project Checkpoints

- 1 Research question approved
- 2 Test plan approved
- 3 Trial data collected
- 4 Graphs/statistics complete
- 5 Final performance claim submitted

Success Criteria

Experimental Design	Variables, controls, and procedure create a fair and repeatable investigation.
Data Quality	Repeated trials are complete, accurate, and organized with units.
Analysis	Statistics, graphs, and kinematics are correct and meaningful.
Conclusion	Final claim is specific, evidence-based, and connected to design or performance decisions.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.

Unit 5 Project Brief

Integrated Aerospace Systems Capstone

Aerospace Systems Capstone: Design, Test & Defend

Challenge Statement: Design, build, test, and defend an integrated aerospace system that solves an approved mission problem using multiple engineering concepts from the course.

Project Overview

In the capstone, your team combines the POE course concepts into one integrated aerospace system. Your solution must connect mechanisms, structures, controls, materials, fabrication, testing, data, and documentation into a final design that can be defended with evidence.

Mission Scenario

Real aerospace projects are systems of systems. A mission may require a structure, a mechanism, a control process, sensors, fabrication, materials decisions, testing, data analysis, and a final technical recommendation. This capstone gives your team the chance to combine those skills into one final engineering solution.

Design Requirements

- Select an approved mission problem such as drone operations support, payload deployment, launch/landing support, rover/ground system, test fixture, inspection system, or aerospace demonstration system.
- Include at least two major subsystem types: mechanism, structure, control system, sensor/programming element, fluid-power element, fabricated component, or tested material/structural feature.
- Create a complete design brief with problem statement, criteria, constraints, stakeholders, and success metrics.
- Develop concepts, compare options with a decision matrix, and justify the selected architecture.
- Build and test a working prototype or high-quality demonstration model.
- Use test data, calculations, graphs, or performance evidence to defend final design decisions.

Constraints

- Project scope must be realistic for the available class time, materials, and tools.
- Team must follow all tool certifications, safety procedures, and approved flight/build/test zones.
- System must be modular enough to identify subsystems and diagnose failures.
- All major design decisions must be documented and justified with evidence.
- Final presentation must include a clear claim, supporting evidence, and next-step recommendations.

Required Engineering Evidence

Design Brief	Problem, stakeholders, criteria, constraints, success metrics, and mission scenario.
System Architecture	Subsystem diagram, inputs, outputs, interfaces, materials, fabrication plan, and responsibilities.
Prototype Evidence	Build photos, CAD/sketch evidence, subsystem tests, troubleshooting log, and revision notes.

Performance Data	Test procedure, trial data, calculations, graphs, failure points, and final performance results.
Final Design Review	Professional presentation that defends the design and explains recommendations.

Project Checkpoints

- 1 Mission proposal approved
- 2 System architecture complete
- 3 Prototype plan approved
- 4 Subsystem tests completed
- 5 Final design review delivered

Success Criteria

Systems Integration	Subsystems work together and are clearly connected to the mission problem.
Engineering Process	Research, concepts, decision matrix, prototype work, testing, and iteration are documented.
Evidence & Analysis	Data and calculations support design claims and recommendations.
Final Communication	Presentation is organized, professional, and strongly supported by engineering evidence.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.